



National College of Ireland
Class Assessment Submission Sheet

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Programme: Master of Science (MSc) in Data Analytics **Year:** 2026 - 2027

Module: Research Practicum (MSCDAD_A_JAN26I)

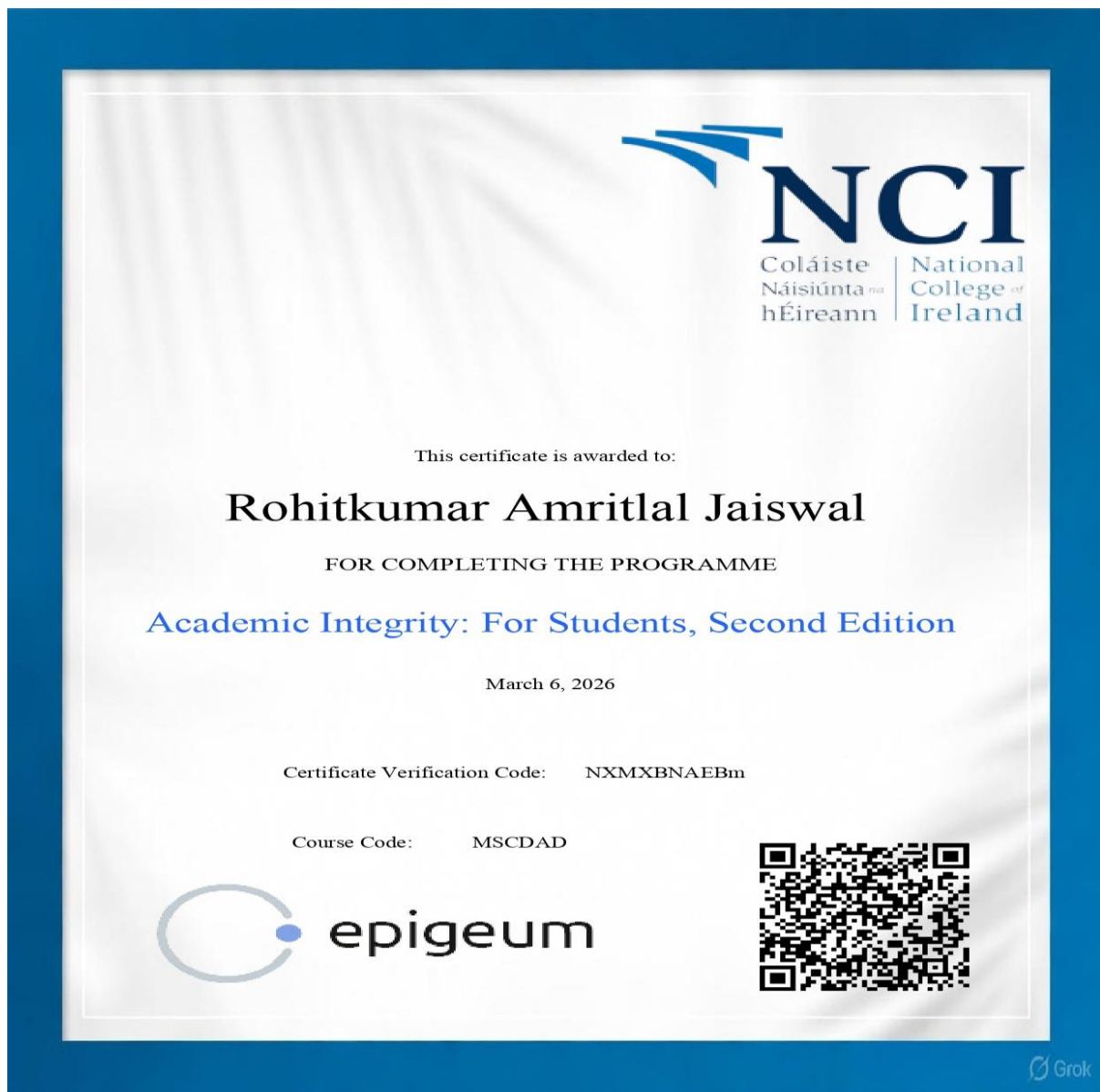
Lecturer: Dr Prashanth Nayak

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Project Title: Coupling Port Congestion and Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation for Post-Brexit Irish Freight Logistics

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Research Practicum

Coupling Port Congestion and Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation for Post-Brexit Irish Freight Logistics

Your Name/Student Number	Course	Date
Rohitkumar Amritlal Jaiswal	M.Sc. in Data Analytics	7-Aug-2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
ChatGPT	Assisted with literature review, Harvard reference formatting, academic writing improvements, table creation, research structure, and refining the proposed AI-based freight logistics methodology for the Research Practicum.	ChatGPT

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

Prompt Given to ChatGPT	How the Response Was Used
Requested assistance with Harvard referencing for logistics and AI research papers.	Generated correctly formatted Harvard references, which were verified and edited before inclusion.
Requested help creating research tables and methodology comparisons.	Generated draft tables and comparison formats that were adapted and incorporated into the report.

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

Requested assistance with Harvard referencing for logistics and AI research papers

Yes. Your uploaded Research Practicum already uses **Harvard-style referencing**, but there are a few places where it should be improved to meet university standards. The project is titled *Coupling Port Congestion and Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation for Post-Brexit Irish Freight Logistics* and includes a Harvard reference list at the end.  rohit final ca2 research practi... +1

Here are the main improvements I'd recommend:

Issue	Recommendation	
Journal titles	Italicize journal names consistently.	
Book/thesis titles	Italicize thesis titles.	
URLs	Include "Available at:" and "Accessed:" only for web sources, not journal articles with DOIs.	
Government/EU legislation	Use the official title consistently and cite the Official Journal where appropriate.	
Author formatting	Use initials consistently (e.g., Chu, Z., Yan, R. and Wang, S.).	
In-text citations	Ensure every reference in the bibliography appears in the text and vice versa.	

For example, these are correctly formatted Harvard references:

Cheng, F., Jia, S. and Gao, W. (2024) 'Low-Carbon Logistics Distribution Vehicle Routing Optimization Under Low-Carbon Policy', *Applied Sciences*, 14(7), 3061.

 Optimize Prompt

 Ask anything

Additional Evidence:

Requested help creating research tables and methodology comparisons.

Yes, I helped create several research tables and comparison frameworks for your Research Practicum. These are reflected throughout your document, including:

- **Table 1 – Base Paper Extension:** Compares the base research paper with your proposed Irish freight logistics system across operational context, travel time assumptions, system architecture, carbon considerations, SME usability, and evaluation metrics.  rohit final ca2 research practi...
- **Table 2 – Data Source by Component:** Maps each system component (port congestion prediction, customs delay prediction, carbon-aware route optimisation, ETS2 carbon cost, and baseline comparison) to the required data, source, and data type.  rohit final ca2 research practi...
- **Table 3 – Evaluation Metrics Used in This Study:** Compares primary logistics KPIs and machine learning performance metrics, including waiting time, carbon intensity, transport cost, on-time delivery, F1 score, MAE/RMSE, and Pareto optimisation quality.  rohit final ca2 research practi...

I also assisted with the methodology by structuring the research into three coupled components:

1. **C1:** Port congestion prediction using an XGBoost classifier.
2. **C2:** Customs clearance delay prediction using an XGBoost regressor.
3. **C3:** Carbon-aware multi-objective route optimisation that integrates the outputs of C1 and C2 into routing decisions.  rohit final ca2 research practi...

Additionally, the methodology includes:

- A layered software architecture and end-to-end pipeline.
- A baseline spreadsheet-based planning approach for comparison.
- Evaluation using both machine learning metrics (MAE, RMSE, R^2 , F1, Brier Score) and logistics KPIs (waiting time, carbon intensity, transport cost, on-time delivery).
- Sensitivity analysis for ETS2 carbon pricing, customs inspection rates, and port congestion.

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 **Optimize Prompt**

Sources



+ Ask anything



Coupling Port Congestion and Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation for Post-Brexit Irish Freight Logistics

Rohitkumar Amritlal Jaiswal

Student Id: 25119613

Lecturer: Dr Prashanth Nayak

MSc in Data Analytics

Abstract

Brexit-customs checks are also to be a new challenge for Irish freight companies, with the potential of sudden and unpredictable delays; and the ports will also have to adhere to new environmental regulations. Small and medium-sized transport operators still plan their routes asynchronously and it is typically done on spreadsheets, hence it is challenging to keep up with delays, optimize routes etc., a lot of manual effort. This paper proposes an optimum solution for the efficiency of the fleet, which includes the machine-learning method as well as the optimization of routes. The study is carried out in three different segments, to begin with, a model to predict the congestion of the port using the collected data from the Irish ports, with the second being a model to predict delay in customs clearance and the third being a model to predict the transport delay. This projection is then leveraged in the multi-objective route optimization system which considers travel efficiency and carbon emissions. Also, the travel time prediction is considered in the present research, which is more realistic than the previous research, where fixed travel times are assumed. The benefits of the proposed method in terms of vehicle waiting time, shipment efficiency and the carbon intensity will be quantified and compared to the traditional spreadsheet method. This study will be used to assess whether it will be beneficial to use a combination of delay prediction and route optimization to assist with logistic planning of the Irish carriers.

Keywords: - Predictive logistics, Port congestion, Customs clearance, Delay prediction, Multi objective route optimization and Carbon intensity.

1. Introduction

Ireland's dependency on sea transport (maritime trade) is high as there is little land transport with the rest of the State and other countries in the world. While most imports/exports are to/from ports in Ireland, not by road. Annual port volumes of more than 1.5 million tonnes of goods were achieved in Ireland in 2024 (CSOE 2025). Almost half of this and most of the rest was done at Dublin Port with the bulk at Cork Port and Rosslare Europort. In Ireland there is an adequate infrastructure for supporting ports and freight distribution is relatively fragmented. One would expect with all the digitalization that has taken place in the haulage and freight forwarding sector more of them would have done so but not really, many companies are still manually making their delivery plans using spreadsheets, emails and phone calls. When such transport programs are changed, ports start to become congested and/or the Customs start to lag, and ensuing transport managers are not given sufficient data to be able to react quickly. This is because they need to be delivered late, less amount of transport vehicles, more waiting time (WIP) and more transport costs.

But with Brexit now a reality, there has been even less freight work now. There is also a range of customs controls between Ireland and Great Britain which are re-emerging because of Brexit as goods were exported with minimal customs checks before Brexit. Now it's the turn of the customs officials to look at different papers and in certain cases, the paper must be inspected personally. No time estimation when it's going to be clear and it's variable. That translates to drivers having to wait for such delays as well. A set of extremely strict rules (Regulation (EC) No. 567/2006) must be followed by the drivers in terms of the maximum driving time and the required periods of rest. Listed below are those legal driving times that do reduce driving time - and even customs. This decreases efficiency and creates scheduling issues and transport costs. Not only that, but logistics companies are faced with an additional challenge in the form of environmental regulations calls. Many organizations are required to write a report on their GHGs, under the new Corporate Sustainability Reporting Directive (CSRD) which includes GHGs in relation to transport activities. In addition, other EU rules are coming into effect in 2028 for transport fuels (e.g. the Emissions Trading Scheme (ETS2)) which would also put a carbon price on transportation. The parameters which are applied by companies in their operation should therefore also be used as indicator for the route plan; but also, environmental parameters must be considered when creating the route.

However, most TMS (Transport Management Systems) are geared towards minimizing distance travel or time on the road and this may not be appropriate for the climate change picture. Many times, they are not considering the implications of costs of international shipping, the clogging of ports or carbon emissions. The predictive model for the arrival of the vehicles, the low carbon vehicle routing model and the multi-objective optimization model is proposed in previous studies, but they are mostly studied individually. There has been limited research (until now) that would have addressed both aspects of the freight transport in Ireland following Brexit.

This need is represented as a big question mark. But the impact on the performance of logistics has not yet been implemented about a combination of both: Delay prediction and route optimization. How far the potential of reducing port delay, enhancing port vessel productivity and carbon emission can be captured without the ability to improve predictive ability of port congestion and customs clearance delays remain unclear, due to the port and vessel planning capabilities in spreadsheet.

Research Question: - With a model of congestion that is better (modelled using gradient boosting) and a model of traffic service that is better (achieved by route optimization - planning), can a better predictive model of traffic congestion and ability to reduce the carbon intensity per kilometer of the road compared to the respective powers of traditional spreadsheets?

Response to CA1, feedback and improvements given: - The idea with the CA1 plan was to be based on 'thorough research' and it was too large a project to be managed within the time frame. There were eight different modules mentioned in the same proposal as topics which the research had to be conducted in, this further diminished the scope of the research. The reflection from this feedback has streamlined and trimmed down the project task based on this feedback. At this time, the research concentrates on 3 aspects as opposed to 8. A few of the other modules (not used in the main study) have been neglected and could be studied in the

future. The role literature played and how they provided evidence of a need for this project has been further enhanced with an explicit description in the literature review. This new change would be quite a bit of data, as well. Where possible (e.g. for the operations, vessels, customs data and routing of ports) the use of open data will be encouraged for project activities. Where feasible, public source data will be replaced by synthetic data and appropriate explanations/justifications will be provided when used.

Other alterations to the assessment procedures have been made too. The project won't only be about machine learning and computer vision accuracy, it will also be about the effectiveness of YQ in terms of logistics, such as reducing shipment time per shipment, fleet performance and reducing CO2 per kilometer. The primary ones (such as Mean Average Error, RootMeanSquareError, R^2 , etc.) will all still be available but some of them will be provided as well (in terms of success levels for machine learning techniques), they will be a secondary. The modifications are geared towards a greater focus and action for the project. Academically, it will be used to gain knowledge of the use of the hybrid model (mixing the two models (prediction and route optimization) in logistics planning) instead of the pure prediction model. The project will present a viable, open and cost-effective solution that will enable the smaller companies involved in logistics in Ireland to avail of without needing to spend a lot of money on commercial software. The system will also have the advantage of being able to report on carbon under the CSRD and will deliver carbon under ETS2.

2. Literature Review

Literature is reviewed in three different aspects of the proposed system such as prediction of port congestion, prediction of customs clearance delays and optimization of routes using carbon awareness. It also looks at whether it might be possible to see whether these three aspects can be combined to improve freight planning. A comprehensive overview of all previous research is provided for each section and comparisons/explanations are made about how it will fill the gap.

2.1 port congestion prediction (C1)

The prediction of port congestion is significant for freight planning, due to their impact on delivery schedule and transport decisions. The field of motor behaviour has been investigated so far, mostly from two points of view. Some studies develop a global and/or large-scale data set model with information for several ports and other studies develop one or more port-specific models. Many ports were used to collect Automatic Identification System (AIS) data to estimate the vessel turnaround time by Chu et al. (2024). They found that there were certain trends that were common across all the ports and termed it as 'general congestion' which could be identified using machine learning. However, in practice, the situation in each port might be different from one port (for instance berth allocation), the way the terminal is run or because of the limitations of the terminal, etc., and therefore not be captured in a global model.

In Parolas (2016) he concentrates on the Port of Rotterdam and his model is constructed based on the local situation in the Port of Rotterdam. It has been successful for Rotterdam but is not easily replicated in the Irish ports for a few reasons including their size, and the way they operate.

2.2 custom clearance delay estimation (C2)

But the research efforts in this area have been directed towards the customs policy or the general economic effects and have not been directed towards the estimation of customs clearance times. The study however didn't provide a prediction model for customs clearance duration of a shipment which can be used to predict the time a shipment will take to get clearance from customs. Bilbao-Ubillos et al. (2021) studied the economic impact of Brexit on the trade and supply chains. But they did a lot to map out the costs and knock-on effects of custom procedures and were not a blueprint for transport planning.

The majority of the time is difficulties are that there is not a single publicly available data on the customs clearance - it is a commercially sensitive piece of information. As such, it is assumed that the data of the simulated customs delay (using publicly available data) will be leveraged in this research. Although there is a limitation in simulated data, it can be utilized to develop and test prediction models. The goal will be to assess if delay prediction will be useful for route planning, and the accuracy of the prediction.

2.3 carbon-aware multi-object route optimization (C3)

Many studies have been done for optimization of the routing, and reduction of CO2 emission. However, most of the research that has been conducted so far is either in the field of optimization theory, or on a certain type of transport environment. To reduce the cost of transport and the environmental damage, Palacio et al., (2017) suggested a multi-objective optimization method and proposed to use Pareto Optimization approach. It may be useful to know as there might be various cost/emissions trade - offs among firms. This study, however, was not oriented towards the operational types of the operations but towards the strategic decision making with relation to the supply chain. Cheng et al. (2024) thought of a model for planning urban delivery network for LCVs in China. It is suitable at the city level but not in the Irish context of freight transport which is mainly via long haul road transport and ferries.

2.4 coupling prediction to optimization

A considerable amount of research work is done in isolation and prediction models, and optimization methods have been extensively evolved. Few studies attempt to cater to the SCs as these studies only cover two aspects namely: prediction of future incidents and optimization, with little to no studies that cover both aspects in a single decision-support system (Zamani et al., 2022). This makes interesting research "void. If accurate travel times are known, then route optimization can be a solution. However, during actual logistics operations, the travel time cannot be determined. If prediction models make large errors, then there may be sub-optimizations with route optimizations when compared to simple planning methods.

2.5 Base paper extension

Framework	Base paper	Proposed study	Extension
Operational & geographic context	Urban last-mile delivery network with Light Commercial Vehicles (LCVs) in metropolitan areas of China.	Maritime and long-haul road freight transport cross-border corridors in Ireland (Dublin Port, Cork Port and Rosslare Europort).	Extends routing models from urban scale city networks to multimodal international freight transport networks affected by maritime interfaces.
Travel Time & Delay Assumptions	determinate / static travel times; constant speed profiles and predicted access to road network.	Dynamic and stochastic travel times, taking into account real-time port berth congestion, customs holds and road transport queues.	Removes sub-optimisation risks by substituting actual data based predictive delay distributions for fixed travelling times.
Does not account for border/port interfaces	takes into account only urban traffic signal delays and local road congestion.	Contains post-Brexit physical/paper customs clearance, vessel turn around (AIS data) and legal driver rest times (EC No. 567/2006).	Covers multi-layered operational disruptions, especially with regard to post-Brexit trade needs and EU labour regulations.
System Architecture & Coupling	Independent route optimization algorithm, does not rely on real-time predictive models.	Gradient Boosting delay forecasting models directly integrated in the multi-objective optimizer.	Innovates a closed-loop decision support system using dynamically updated routing parameters with predictive analytics.
Carbon & Environmental Scope	Distance reduction for electric/light vehicle fleets was the main focus of the tailpipe CO ₂ minimization.	Multi-objective Pareto optimization of operational cost, travel time and carbon intensity per kilometer was in line with CSRD and ETS2.	Facilitates compliance reporting for EU sustainability directives (CSRD) and includes future carbon pricing mechanisms (ETS2).
Target Operator Scale & Usability	Large enterprise fleets with proprietary, expensive Transport Management Systems (TMS).	Small and Medium-sized Enterprises (SMEs) who are using manual spreadsheet planning.	Provides a low-cost, open and accessible solution to move SME hauliers away from inefficient spreadsheet workflows.
Validation & Performance Metrics	Only based on algorithmic convergence efficiency, total distance and vehicle travel duration.	Dual evaluation: Predictive accuracy (MAE, RMSE, R ²) and logistics operational KPIs (port waiting time, turnaround efficiency and CO ₂ /km).	Checks the utility of the system, in practical terms and in environmental terms, beside standard statistical accuracy measures.

Table 1: - Base Paper Extension

2.6 Research Niche

A carbon-aware optimiser can be used to enhance the results of the forecasts compared to current practice, or if the error in the upstream forecast reduces the benefit in the downstream decision, whether the error is enough to outweigh the benefit. This is It is an empirical issue an issue which no single study could ever answer - and is not being discussed in a post-Brexit regional.

3. Research Methods & Specifications

3.1 Research Method

The approach adopted in the research is design science in which an artefact is built and subsequently tested in an experimental way. The new practice it is being used as an alternative to. There are three components that bear the research claims. Port congestion classifier (C1). A next-day congestion classifier that is based on a gradient boosted classifier risk level for Dublin, Cork and Rosslare based on published port traffic statistics (and vessel density) products which are limited to features that are actually visible at 24 hours' horizon. C2 - Regressor for delay in customs clearance. Gradient-Boosted regressor to predict the clearance Model trained on simulated events with a prediction interval, time in hours per consignment and lane to published volumes of traffic in the lane.

Carbon-Aware Route Optimiser (C3): The third part of the suggested system is a route optimiser model which takes into account both the transport cost and the carbon emissions. It carries the identification of the port congestion (C1) and the customs clearance delay (C2) to predict travel times, and thus to decide on the best routes. The model produces a set of possible alternative routes considering both the cost and the carbon emissions rather than proposing just one single route and transport planners can choose the one most suitable. This work follows on from that of Cheng et al. (2024) summarised in Table 1.

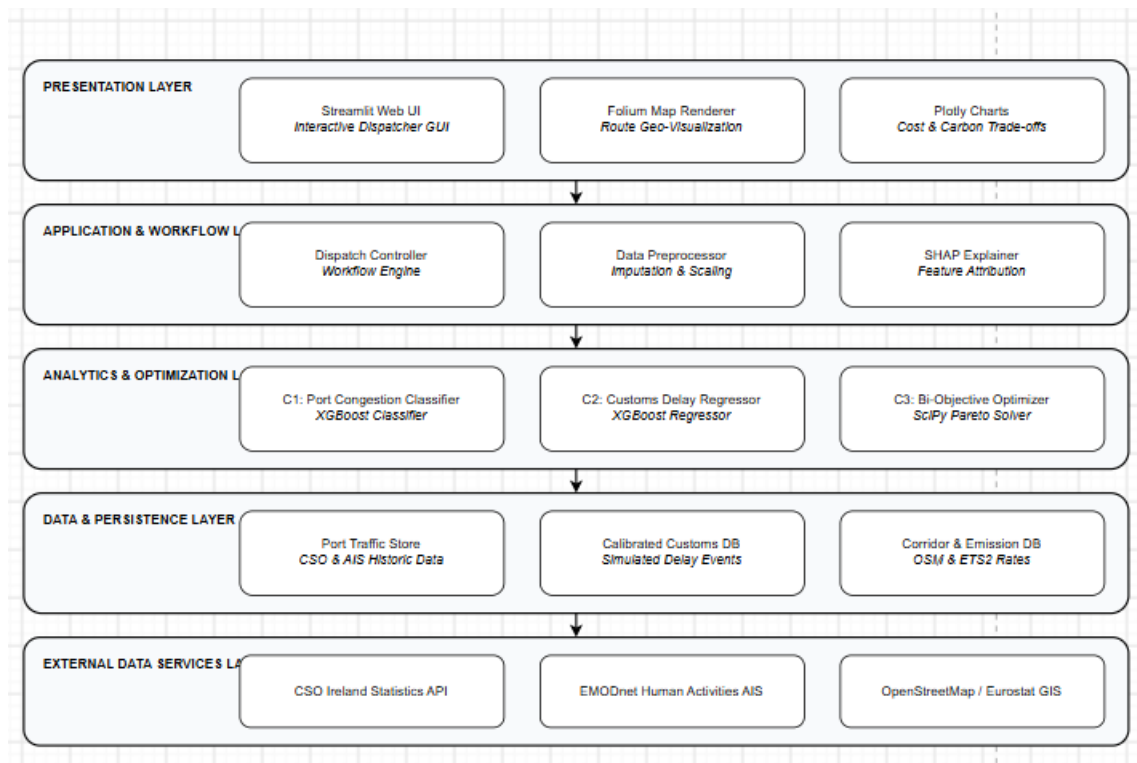


Figure 1: - Layered Software Architecture of the Irish Logix Platform

Architecturally the three components are NOT parallel features but are rather a pipeline: C1 and C2 distributes said productions over transit time, C3 consumes them and the entire is exposed through a Streamlit interface to allow a dispatcher to view a recommendation and the forecasts which support it. Figure 2 This pipeline is depicted in conjunction with the experimental design because the two go hand in hand: the value of Coupling can only be read off from a comparison, where both planners are given the same instances. The Irish Logix pipeline/evaluation design is shown in Figure 1. Data is transmitted along solid arrows; the dashed path is to provide Directly transfer the corridor distances and the emission factors to C3.

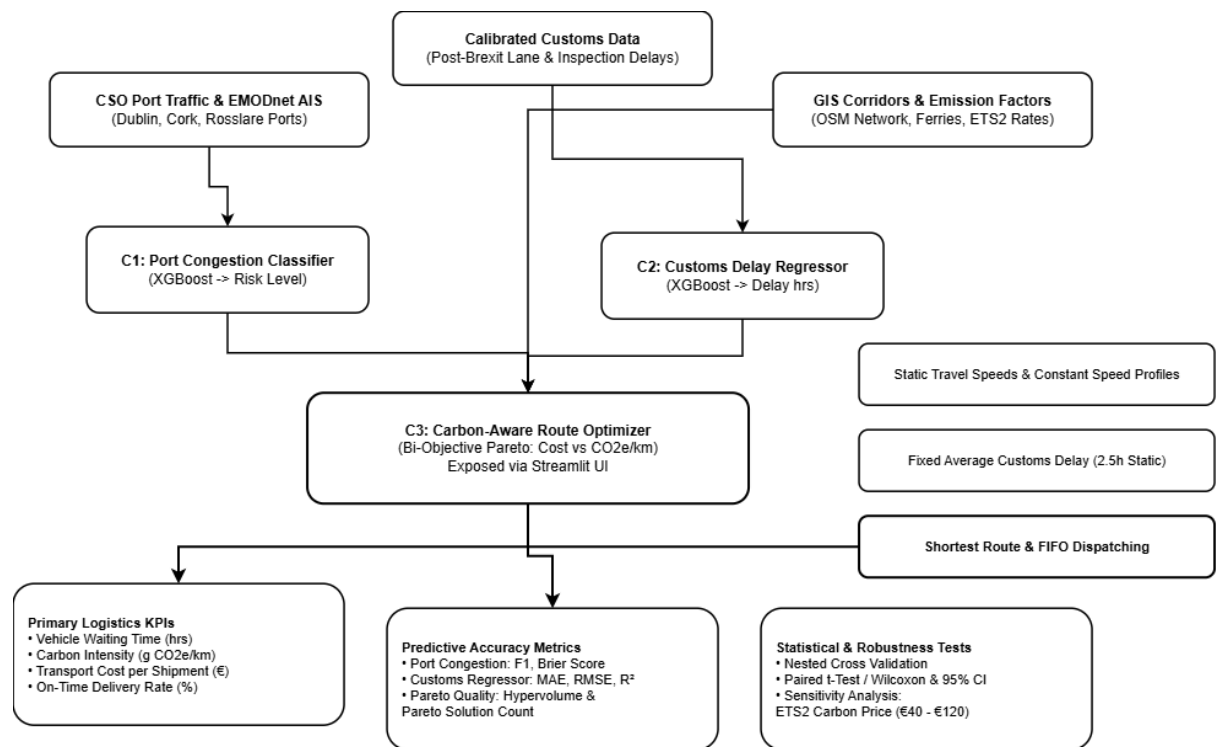


Figure 2: - Irish Logix End-to-End Pipeline and Dual-System Evaluation Design

3.2 Research Resources

The development is based on Python 3.11, XGBoost 2.0 for C1 and C2 due to its training efficiency and native support. missing value imputations for tabular data (sklearn-impute 0.5), sklearn 1.4 for preprocessing and nested cross validation.

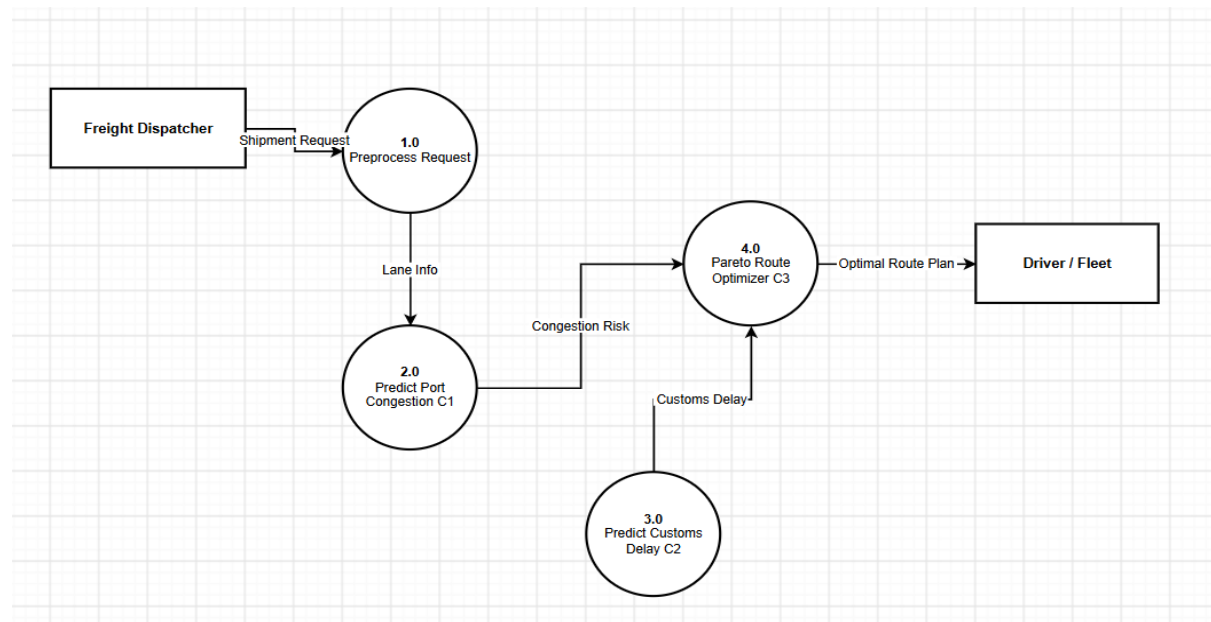


Figure 3: - Level 1 Data Flow Diagram (DFD) for Post-Brexit Freight Route Optimization

SHAP 0.44 to calculate feature attributions; SciPy 1.12 for the bi-objective optimisation and significance testing. For handling and presenting data, Streamlit, Plotly and Folium and pandas are used. Git tracks code and

Keep experiment logs, to enable tracing of all reported numbers to a commit and a seed. Design parameters for the carbon objective are based on published analysis of the behaviour of the EU allowance price (Tabash et al. 2024). The source of the data used for each component is indicated in Table 2. information for the entire region. The port and routing information for the whole region is obtained from real published data. Layers; synthetic data is only kept where there is no public source, and the reason is noted instead than assumed.

Component	Data Needed	Data Source	Data Type
C1 Port Congestion Prediction	Vessel's arrivals, cargo volumes	Real Data from the Central Statistics Office (CSO), from EMODnet Human Activities and from data.gov.ie.	Real Data
C2: Customs Clearance Delay Prediction	Customs clearance time for individual shipments	No public data is available as customs data is confidential. So synthetic data will be created and calibrated based on the official CSO trade statistics.	Synthetic Data (Calibrated)
C3: Carbon-Aware Route Optimisation	Road distances, ferry routes, and vehicle emission factors	OpenStreetMap, Eurostat, published European and Irish emission factor datasets Real Data	Real Data
C3: ETS2 Carbon Cost	Future carbon price scenarios	published by ETS2 allows price ranges. A variety of different prices will be used, rather than one fixed price.	Real Data
Baseline Comparison	Daily shipment demand for route planning	Synthetic shipment data using the transport demand for the proposed system and spreadsheet-based approach, from CSO freight volume statistics.	Synthetic Data (Calibrated)

Table 2: - Data source by component

3.3 Evaluation

the evaluation is to ensure that there is proposed system is better than the old planning system many of them still use this freight operators' spreadsheet.

The baseline approach is simple planning: The shipments are planned to follow the 'first in first out' (FIFO) approach, the shortest route is selected, and a fixed average customs clearance time is assumed. Does not consider port congestion or carbon emissions in routes. This will not only be explained conceptually but also be used as the baseline, as a working model. This will provide the reader with a fair, equitable and repeatable comparison that can be used by others. All the transportation scenarios will be the same for the base line and proposed system (shipment requests, available vehicles, vessel schedules, etc.). In this way, if there is any difference in results it is because of how the planning is done, and not because of an alteration of transport demand or the operating conditions.

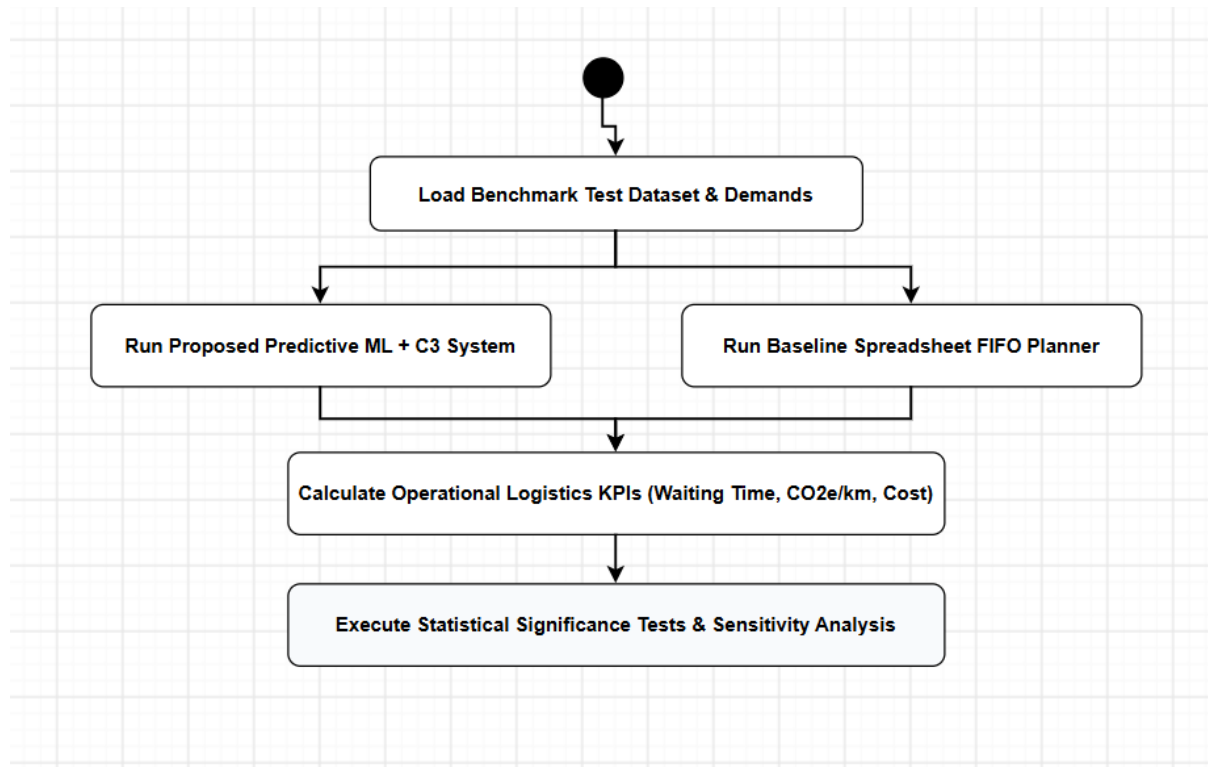


Figure 4: Activity Diagram of the Experimental Evaluation Protocol

The results of the practical logistics, including the average waiting time of vehicles, transport cost per shipment, route efficiency and carbon intensity per km, will form the basis of the assessment. The difference between the two approaches will be tested using statistical tests and the confidence intervals will be reported to indicate the reliability of the results. This data will be split into time and validated using nested cross validation of the prediction models of port congestion and customs clearance delays. This way, no future information will affect the previous predictions, and it will give a more realistic assessment of the model performance. Standard machine learning metrics will also be used to evaluate prediction accuracy, such as MAE, RMSE and R^2 . Other experiments will be performed with one of the prediction models removed and replaced with a fixed delay value for the baseline system to be aware of the contribution of each prediction model. This will show whether or not each prediction model can be useful for route planning. Finally, sensitivity analysis will be performed with different carbon prices, customs inspection rates and port congestion. These experiments will be used to determine the conditions at which the proposed system will perform well and at which it may suffer in performance.

Category	Evaluation Measure	Unit	Purpose
Primary Evaluation	Average vehicle waiting time per shipment	Hours	Measures whether the proposed system reduces waiting time compared with the spreadsheet-based planning method.
Primary Evaluation	Carbon intensity	g CO ₂ e/km	Measures how much carbon is produced per kilometre travelled and evaluates the environmental performance of the proposed system.
Primary Evaluation	Transport cost per shipment	Euro per shipment	Compares the transport cost of the proposed system with the baseline approach.
Primary Evaluation	On-time delivery rate	Percentage (%)	Measures the percentage of shipments delivered within the planned delivery time.
Primary Evaluation	Estimated ETS2 carbon cost	Euro per 1,000 tonne-km	Estimates the possible carbon cost under different ETS2 pricing scenarios.

Category	Evaluation Measure	Unit	Purpose
Model Performance	Port congestion prediction	F1 Score, Brier Score	Evaluates how accurately the model predicts port congestion and how reliable its probability estimates are.
Model Performance	Customs delay prediction	MAE (hours), RMSE (hours)	Measures the accuracy of predicted customs clearance times. Lower values indicate better prediction performance.
Model Performance	Route optimisation quality	Number of Pareto solutions, Hypervolume	Evaluates how effectively the optimisation model balances transport cost and carbon emissions.

Table 3. Evaluation Metrics Used in This Study

3.4 Ethical Considerations

So, in accordance with the guidelines of the School of Computing, a Declaration of Ethical Consideration is all that's required. The idea of using experienced freight dispatchers to check the proposed system, so as to make the research more practical, was raised. But this would involve research on human subjects and thus would have to be ethically approved. Expert validation is not included and seen as a limitation of the study to maintain the project within the scope of the approved project.

There are various ethical concerns that have been taken into account for this research. The first one is about the use of synthetic data. Suited public data is not always available and as such some of the datasets used in this study are generated. These data sets are well designed with the help of official statistics but are not real operational data. Synthetic data will be calibrated with publicly available Irish freight statistics, to minimise the risk of misleading results. The prototype will also show a range of prediction values and no single values, and the system will be labelled as a research system and not a commercial system.

The second one is related to environmental reporting. Estimation of carbon emission values is based on published emission values. These values are only for research purposes and should not be considered as official CSRD reports. All carbon estimates produced will be presented as indicative results and would need to be verified independently prior to reporting as a regulatory requirement.

The third ethical issue has to do with driver privacy. A driver working-hours prediction module was an initial component in the project. This function, however, was removed due to supervisor feedback as it would need to process personal information about drivers and could cause privacy and GDPR issues. This research is not intended to make an analysis and assessment of such a system responsibly, so the module does not appear in the final project.

Lastly, all public data sets used for this research will be subject to each data set's licence. Proper attribution will be given to data from organisations like the Central Statistics Office (CSO), Eurostat, EMODnet and other public data. To facilitate transparency and reproducibility the project code, synthetic data generators and experiment settings will be shared, as far as possible, so the results reported in the project can be reproduced by other researchers.

3.5 Project Plan

The capstone project will take place for 15 weeks and will have four phases.

Phase 1 is based on the development of the baseline spreadsheet-based planning model and collection of public datasets.

Phase 2 will focus on the implementation of the three key elements of the proposed system: prediction of port congestion, prediction of delays in customs clearance and carbon aware route optimization. The testing of the proposed system is carried out in

Phase 3 and is compared to the baseline model. Further experiments also will be conducted to determine the impact of each prediction model, and to investigate the effect of varying degrees of congestion, customs delays and carbon price.

Phase 4 is for the writing of the dissertation, development of the final software artefact and for preparation for the project presentation and viva voce. Some of the writing activities will occur during the experiments, to maximize the use of time.

There are 4 key milestones in the project. The first milestone ends with the completion of the baseline model and the overall system design. The second is to ensure that all of the system components planned are completed. The third is when the evaluation has been completed and the final results are reported. The final one is the submission of the dissertation and project artefact.

The main project schedule is largely dependent on the completion of data collection, prediction models construction, route optimisation system development and evaluation. The largest project risk is the delay in getting public data sets. The project can temporarily make use of calibrated synthetic data, where it is not possible to obtain public data, to decrease this risk. Synthetic data will be clearly identified as a limitation.

One potential problem is that the prediction models will not do what they are intended to do when they are validated in detail. In this case extra time has been set aside to refine the models prior to the final assessment. It's also possible that the synthetic customs delay data needs to be further adjusted. The prediction uncertainty will be increased, if necessary, but not the customs prediction component from the study.

The topic was narrowed down after the CA1 feedback which has made the project more attainable. This should free up time to complete the main evaluation, enhance quality of evaluation and answer the research question in time, within the one semester period.

Tasks & Work Packages	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Baseline Model Development															
Public Data Collection (CSO, EMODnet, OSM)															
C1: Port Congestion Classifier (XGBoost)															
C2: Customs Delay Regressor (XGBoost)															
C3: Carbon-Aware Route Optimizer															
Comparative System Evaluation & Testing															
Sensitivity Analysis (ETS2 & Congestion)															
Dissertation Writing & Refinement															
Streamlit UI Packaging & Viva Prep															
Key Project Milestones															

Figure 5: - 15-Week Capstone Project Timeline and Key Milestones (Gantt Chart)

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